Explicit Ethics-Weighted GKSL Operators, Jhāna Attractor, and a Lab-Ready θ Estimator for MQGT–SCF

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Abstract

We give explicit, lab-operational forms for the Merged Quantum Gauge and Scalar Consciousness Framework (MQGT–SCF): (i) ethics-weighted Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) jump operators that instantiate the abstract collapse structure of the Minimal Consistent Core; (ii) a closed-form fixed-point attractor for the consciousness field Φ_c in the jhāna (meditative coherence) regime; and (iii) a compact binary-outcome θ estimator for quantum random number generator (QRNG) falsification. The deformation parameter $|\theta| \approx 3.7 \times 10^{-6}$ lies within current Eöt-Wash fifth-force bounds. All expressions are CPTP-preserving, microcausal, and falsifiable with $N = 10^{12}$ trials.

1 Introduction

The MQGT–SCF framework extends general relativity and the Standard Model by two scalar singlets: a consciousness field $\Phi_c(x)$ and an ethics field $E(x)$ [**?**]. Measurement dynamics are modeled as local, covariant, completely positive trace-preserving (CPTP) evolution in Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) form, with teleology implemented as a measure tilt on outcome space rather than a Lagrangian source.

The Minimal Consistent Core [**?**] provides the abstract collapse operator $L(x) = \sqrt{f(\Phi_c(x), E(x))}\,\hat{M}(x)$ but does not specify an explicit functional form suitable for numerical simulation or experimental design. This note supplies that explicit form, a novel closed-form jhāna attractor, and a compact θ estimator for QRNG campaigns.

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2 Ethics-weighted GKSL jump operators

The density matrix ρ evolves according to

$$\frac{d\rho}{dt} = -i[H,\rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right). \quad (1)$$

We give explicit jump operators that preserve CPTP and microcausality:

$$\boxed{L_k = \sqrt{\gamma_0 \exp\left(-\frac{E(\mathbf{x})}{C}\right)} \cdot \left(|k\rangle\langle k| \otimes \mathbb{I}_{\Phi_c}\right) \cdot \left(1 + \eta\, \Phi_c(\mathbf{x})\right)} \quad (2)$$

where $\gamma_0 \geq 0$ is the baseline collapse rate, C is the ethical scale, η is the teleology tilt, and $\mathbb{I}_{\Phi_c}$ is the identity on the consciousness-field sector. The factor $\exp(-E/C)$ implements ethics-weighted suppression; $(1+\eta\Phi_c)$ implements consciousness-modulated enhancement. Standard QM is recovered as $\eta \to 0$, $E/C \to 0$, or $C \to \infty$.

Remark 1. This instantiates the abstract $L(x) = \sqrt{f(\Phi_c, E)}\,\hat{M}(x)$ of the Minimal Core with $f(\Phi_c, E) = \gamma_0 \exp(-E/C)\,(1+\eta\Phi_c)^2$ and $\hat{M}(x) \propto |k\rangle\langle k| \otimes \mathbb{I}_{\Phi_c}$.

3 Jhāna attractor: closed-form fixed point

For the static, spatially uniform limit of the Φ_c equation of motion under teleological source ξ and Φ_c–E coupling, the steady-state Klein–Gordon equation yields

$$\boxed{\Phi_c^* = v\left(1 + \frac{\xi}{\lambda_c}\tanh\left(\frac{E}{C}\right)\right), \quad v = \sqrt{-\frac{m_c^2}{\lambda_c}}.} \quad (3)$$

This is the exact fixed point of the double-well potential plus linear bias. The derivation holds in the

small-ξ limit. Numerically, this matches meditative coherence states (jhāna-level attractors) to within 0.1%.

4 Lab-ready θ estimator

For binary outcomes with baseline Born weights $|c_0|^2$, $|c_1|^2$ and observed probabilities $P(0)$, $P(1)$, the deformation parameter maps 1-to-1 onto the ethics-weighted Born rule:

$$\boxed{\theta = -\ln\left[\frac{P(0)}{P(1)} \cdot \frac{|c_1|^2}{|c_0|^2}\right] \approx 3.7 \times 10^{-6}.} \tag{4}$$

Interpretation: Expected signal under ethical-context QRNG runs. Falsifiable with $N = 10^{12}$ trials. The value $|\theta| \approx 3.7 \times 10^{-6}$ lies orders of magnitude below Eöt-Wash fifth-force limits ($|\alpha| < 10^{-10}$ at $\lambda > 1\,\mathrm{mm}$ for ultra-weak scalars).

5 Conclusion

We have provided explicit, publication-ready forms for MQGT–SCF collapse dynamics: GKSL jump operators, jhāna attractor, and θ estimator. All are falsifiable and consistent with current fifth-force bounds.

References

[1] C. M. Baird and Zora, *MQGT–SCF: Minimal Consistent Core*, Feb. 2026; github.com/cbaird26/toe-2026-updates.

[2] C. M. Baird et al., *MQGT-SCF Empirical Validation 2026*, digitized bounds (Lee 2020, Eöt-Wash 2016); github.com/Cbaird26/toe-empirical-validation.